

Title

Vascular Effects of Combined Mechanical and Photobiomodulatory Periodontal Treatment: Study Design and Methodology

Abstract

Periodontal inflammation is implicated in systemic vascular pathology through multiple inflammatory pathways. While periodontal intervention improves local tissue health and endothelial function, its effects on larger arterial hemodynamics remain undercharacterized. This randomized trial will compare two evidence-based periodontal treatment approaches regarding their impact on common carotid artery hemodynamic properties. Enrollment targets 40 adults with active periodontal inflammation requiring therapeutic intervention. One arm receives conventional subgingival debridement; the parallel arm receives identical debridement plus adjunctive photobiomodulation therapy. Serial vascular assessments at baseline, immediate post-treatment, and 20-week intervals will measure arterial wall tension dynamics, blood rheology, and elastic properties. Clinical periodontal parameters will be monitored throughout. This study addresses whether combining mechanical and light-based modalities produces superior vascular adaptation compared to mechanical therapy alone.